DevOps Fundamentals

✦ topper notes • v2 • dense book style

INTERVIEW HOT ⭐
Revise CI/CD daily!
must ↓

Today's Learning Objectives 6 POINTS ✓

- ◯ 1. Understand DevOps and its lifecycle.
- ◯ 2. Understand SDLC.
- ◯ 3. Learn CI, Continuous Delivery and Continuous Deployment.
- ◯ 4. Understand the DevOps engineer's responsibilities.
- ◯ 5. Complete your first practical exercise.
- ◯ 6. Prepare for interview questions.

1 Part 1 – What is DevOps?

1. Definition core · must

DevOps is a combination of Development (Dev) and Operations (Ops).

It is a culture, set of practices and tools that help development and operations teams work together to deliver software faster, more reliably and continuously.

Not a tool, it's CULTURE ✨
Mindset > Tools!

Let's understand with an example.

Imagine a company develops an e-commerce website.

1. – Developers write application code.
2. – Testers test the application.
3. – Operations teams deploy the application to servers.
4. – DevOps engineers automate building, testing, deployment and monitoring.

❌ *Without DevOps*
Teams work separately, causing delays and communication problems.

→

✅ *With DevOps*
We automate and integrate the entire process.

2. DevOps lifecycle ♾ 8 stages

Stage	Purpose	Tools
Plan	Plan requirements	Jira
Code	Write code	VS Code, Git
Build	Compile app	Maven
Test	Test quality	Junit, SonarQube
Release	Prepare release	Jenkins
Deploy	Deploy app	Docker, K8s
Operate	Manage infra	Linux, Ansible
Monitor	Monitor perf	Prometheus

MUST KNOW★

! **Important: DevOps is not a single tool. It is a complete working approach.**

2 Part 2 – Understand SDLC

SDLC stands for Software Development Life Cycle. It is the process followed to develop software.

The main stages are:

1. Requirement gathering
2. Planning and design
3. Development
4. Testing
5. Deployment
6. Maintenance

Difference between SDLC and DevOps: ⚖

SDLC	DevOps
Describes overall process of developing software.	Focuses on collaboration, automation and continuous delivery.

3 Part 3 – Understand CI/CD ⭐ INTERVIEW HOT TOPIC

Most important for DevOps interviews.

1. Continuous Integration (CI)

Developers frequently push code into shared repo like GitHub.

An automated pipeline:

- Fetches the code.
- Builds the app.
- Runs automated tests.
- Reports errors.

Example 💡
Jenkins builds app whenever new code is pushed to GitHub.

2. Continuous Delivery

Prepared for deployment. May need manual approval.

needs manual 👍

3. Continuous Deployment

Automatically deployed without manual approval.

auto 🚀

Concept vs Meaning 👇

Concept	Meaning
CI	Automatically integrate, build and test
Continuous Delivery	Keep software ready for release
Continuous Deployment	Auto deploy successful changes

4 Part 4 – Real DevOps Workflow

Imagine developer updates an app. Process works like:

1. 💻 Developer writes code — VS Code
2. 📤 Code pushed to GitHub — Git / GitHub
3. ⚙ Jenkins starts pipeline — Continuous Integration
4. 🔨 Application is built and tested — Maven / Testing
5. 📦 Application packaged — Docker / Podman
6. ☸ Application deployed — Kubernetes
7. 📊 Application monitored — Prometheus / Grafana

🎉 *Day 1 Done! → Practical Next*
Keep momentum 🚀 • topper complete

handwritten with ❤ • topper notes • open book spread • left 01 / right 02